# BOREL FRACTIONAL CHROMATIC NUMBERS FROM BERNOULLI MEASURABLE INDEPENDENCE IN SCHREIER GRAPHS

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*Editorial note.* Complete original mathematical body and bibliography retained. Authors retain responsibility for mathematics.

*Editorial note.* Original countable group, finite F, free Borel action and Bernoulli product measure definitions retained. The Bernoulli-shift equality and general-action inequality are one theorem; no equality for every action is added.

*Allowed standard prerequisites (editorial).*

- Author-cited Borel coloring, clopen approximation and measurable independent-set inputs. Original cited prior inputs remain cited; all source-local constructions and complete proofs are retained.

## 1. Definitions and results

All graphs in this paper are undirected and simple. Recall that for a graph $G$, a subset $I \subseteq V(G)$ is *$G$-independent* if no two vertices in $I$ are adjacent in $G$. The *chromatic number* of $G$, denoted by $\chi(G)$, is the least $\ell \in \mathbb{N}$ such that there exist $G$-independent sets $I_1, \dots, I_\ell$ whose union is $V(G)$. (If no such $\ell$ exists, we set $\chi(G) \coloneqq \infty$.) The sequence $I_1, \dots, I_\ell$ is called an *$\ell$-coloring* of $G$, where we think of the vertices in $I_i$ as being assigned the color $i$.

Fractional coloring is a well-studied relaxation of graph coloring. For an introduction to this topic, see the book [15] by Scheinerman and Ullman. Given $k \in \mathbb{N}$, the *$k$-fold chromatic number* of $G$, denoted by $\chi^k(G)$, is the least $\ell \in \mathbb{N}$ such that there are $G$-independent sets $I_1, \dots, I_\ell$ which cover every vertex of $G$ at least $k$ times (such a sequence $I_1, \dots, I_\ell$ is called a *$k$-fold $\ell$-coloring*). Note that the sets $I_1, \dots, I_\ell$ need not be distinct. In particular, if $I_1, \dots, I_{\chi(G)}$ is a $\chi(G)$-coloring of $G$, then, by repeating each set $k$ times, we obtain a $k$-fold $k\chi(G)$-coloring, which shows that

$$\chi^k(G) \leqslant k\chi(G) \quad \text{for all } k.$$

This inequality can be strict; for example, the 5-cycle $C_5$ satisfies $\chi(C_5) = 3$ but $\chi^2(C_5) = 5$. It is therefore natural to define the *fractional chromatic number* $\chi^*(G)$ of $G$ by the formula

$$\chi^*(G) \coloneqq \inf_{k \geqslant 1} \frac{\chi^k(G)}{k}.$$

In this note we investigate fractional colorings from the standpoint of Borel combinatorics. For a general overview of Borel combinatorics, see the surveys [7] by Kechris and Marks and [12] by Pikhurko. The study of fractional colorings in this setting was initiated by Meehan [11]; see also [7, § 8.6]. We say that a graph $G$ is *Borel* if $V(G)$ is a standard Borel space and $E(G)$ is a Borel subset of $V(G) \times V(G)$. The *Borel chromatic number* $\chi_{\mathrm{B}}(G)$ of $G$ is the least $\ell \in \mathbb{N}$ such that there exist Borel $G$-independent sets $I_1, \dots, I_\ell$ whose union is $V(G)$. The *Borel $k$-fold chromatic number* $\chi^k_{\mathrm{B}}(G)$ is defined analogously, and the *Borel fractional chromatic number* $\chi^*_{\mathrm{B}}(G)$ is

$$\chi^*_{\mathrm{B}}(G) \coloneqq \inf_{k \geqslant 1} \frac{\chi^k_{\mathrm{B}}(G)}{k}.$$

A particularly important class of Borel graphs are Schreier graphs of group actions. Let $\Gamma$ be a countable group with identity element $\mathbf{1}$ and let $F \subseteq \Gamma$ be a finite subset. The *Cayley graph* $G(\Gamma, F)$ of $\Gamma$ is the graph with vertex set $\Gamma$ in which two distinct group elements $\gamma$, $\delta$ are adjacent if and only if $\gamma = \sigma\delta$ for some $\sigma \in F \cup F^{-1}$. This definition can be extended as follows. Let $\Gamma \curvearrowright X$ be a Borel action of $\Gamma$ on a standard Borel space $X$. The action $\Gamma \curvearrowright X$ is *free* if

$$\gamma \cdot x \neq x \quad \text{for all } \ x \in X \quad \text{and} \quad \mathbf{1} \neq \gamma \in \Gamma.$$

The *Schreier graph* $G(X, F)$ of an action $\Gamma \curvearrowright X$ is the graph with vertex set $X$ in which two distinct points $x$, $y \in X$ are adjacent if and only if $y = \sigma \cdot x$ for some $\sigma \in F \cup F^{-1}$. Note that the Cayley graph $G(\Gamma, F)$ is a special case of this construction corresponding to the left multiplication action $\Gamma \curvearrowright \Gamma$. More generally,

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when the action $\Gamma \curvearrowright X$ is free, $G(X, F)$ is obtained by putting a copy of the Cayley graph $G(\Gamma, F)$ onto each orbit.

A crucial example of a Borel action is the *(Bernoulli) shift* $\Gamma \curvearrowright 2^\Gamma$, given by the formula

$$(\gamma \cdot x)(\delta) \coloneqq x(\delta\gamma) \quad \text{for all} \quad x \colon \Gamma \to 2 \quad \text{and} \quad \gamma,\ \delta \in \Gamma.$$

We use $\beta$ to denote the "coin flip" probability measure on $2^\Gamma$, obtained as the product of countably many copies of the uniform probability measure on $2 = \{0, 1\}$. Note that $\beta$ is invariant under the shift action. The *free part* of $2^\Gamma$, denoted by $\mathrm{Free}(2^\Gamma)$, is the set of all points $x \in 2^\Gamma$ with trivial stabilizer. In other words, $\mathrm{Free}(2^\Gamma)$ is the largest subspace of $2^\Gamma$ on which the shift action is free. It is easy to see that the shift action $\Gamma \curvearrowright 2^\Gamma$ is free $\beta$-almost everywhere, i.e., $\beta(\mathrm{Free}(2^\Gamma)) = 1$.

Let $G$ be a Borel graph and let $\mu$ be a probability (Borel) measure on $V(G)$. The *$\mu$-independence number* of $G$ is the quantity $\alpha_\mu(G) \coloneqq \sup_I \mu(I)$, where the supremum is taken over all $\mu$-measurable $G$-independent subsets $I \subseteq V(G)$. Note that if $I_1, \dots, I_\ell$ is a Borel $k$-fold $\ell$-coloring of $G$, then

$$\ell\alpha_\mu(G) \geqslant \mu(I_1) + \cdots + \mu(I_\ell) \geqslant k,$$

which implies $\chi^*_{\mathrm{B}}(G) \geqslant 1/\alpha_\mu(G)$. Our main result is a matching upper bound for Schreier graphs:

**Theorem 1.1.** *Let $\Gamma$ be a countable group and let $F \subseteq \Gamma$ be a finite set. If $\Gamma \curvearrowright X$ is a free Borel action on a standard Borel space, then*

$$\chi^*_{\mathrm{B}}(G(X, F)) \leqslant \frac{1}{\alpha_\beta\left(G\left(\mathrm{Free}\left(2^\Gamma\right), F\right)\right)}. \tag{1}$$

*In particular,*

$$\chi^*_{\mathrm{B}}\left(G\left(\mathrm{Free}\left(2^\Gamma\right), F\right)\right) = \frac{1}{\alpha_\beta\left(G\left(\mathrm{Free}\left(2^\Gamma\right), F\right)\right)}. \tag{2}$$

While (2) is a special case of (1), it is possible to deduce (1) from (2) using a theorem of Seward and Tucker–Drob [14], which asserts that every free Borel action of $\Gamma$ admits a Borel $\Gamma$-equivariant map to $\mathrm{Free}(2^\Gamma)$. Nevertheless, we will give a simple direct proof of (1) in § 2.

An interesting feature of Theorem 1.1 is that it establishes a precise relationship between a *Borel* parameter $\chi^*_{\mathrm{B}}$ and a *measurable* parameter $\alpha_\beta$. We find this somewhat surprising, since ignoring sets of measure 0 usually significantly reduces the difficulty of problems in Borel combinatorics. For instance, given a Borel graph $G$ and a probability measure $\mu$ on $V(G)$, one can consider the *$\mu$-measurable chromatic number* $\chi_\mu(G)$, i.e., the least $\ell \in \mathbb{N}$ such that there exist $\mu$-measurable $G$-independent sets $I_1, \dots, I_\ell$ whose union is $V(G)$. By definition, $\chi_\mu(G) \leqslant \chi_{\mathrm{B}}(G)$, and it is often the case that this inequality is strict—see [7, § 6] for a number of examples. By contrast, as an immediate consequence of Theorem 1.1 we obtain the opposite inequality $\chi^*_{\mathrm{B}}(G) \leqslant \chi_\beta(G)$, where $G$ is the Schreier graph of the free part of the shift:

**Corollary 1.2.** *Let $\Gamma$ be a countable group and let $F \subseteq \Gamma$ be a finite set. Set $G \coloneqq G(\mathrm{Free}(2^\Gamma), F)$. Then $\chi^*_{\mathrm{B}}(G) \leqslant \chi_\beta(G)$.*

*Proof.* Follows from Theorem 1.1 and the inequality $\alpha_\beta(G) \geqslant 1/\chi_\beta(G)$. □

As a concrete application of Theorem 1.1, consider the free group case. For $n \geqslant 1$, let $\mathbb{F}_n$ be the free group of rank $n$ generated freely by elements $\sigma_1, \dots, \sigma_n$ and let $G_n$ denote the Schreier graph of the free part of the shift action $\mathbb{F}_n \curvearrowright 2^{\mathbb{F}_n}$ with respect to the set $\{\sigma_1, \dots, \sigma_n\}$. Then every connected component of $G_n$ is an (infinite) $2n$-regular tree. In particular, the chromatic number of $G_n$ is 2. On the other hand, Marks [10] proved that $\chi_{\mathrm{B}}(G_n) = 2n + 1$. Meehan inquired where between these two extremes the Borel fractional chromatic number of $G_n$ lies:

**Question 1.3** ([11, Question 4.6.3]; see also [7, Problem 8.17])**.** What is the Borel fractional chromatic number of $G_n$? Is it always equal to 2?

Using Theorem 1.1 together with some known results we asymptotically determine $\chi^*_{\mathrm{B}}(G_n)$ (and, in particular, give a negative answer to the second part of Question 1.3):

**Corollary 1.4.** *For all $n \geqslant 1$, we have*

$$\chi^*_{\mathrm{B}}(G_n) = (2 + o(1))\frac{n}{\log n},$$

*where $o(1)$ denotes a function of $n$ that approaches 0 as $n \to \infty$.*

In other words, the Borel fractional chromatic number of $G_n$ is less than its ordinary Borel chromatic number roughly by a factor of $\log n$. We present the derivation of Corollary 1.4 in § 3.

## 2. Proof of Theorem 1.1

We shall use the following theorem of Kechris, Solecki, and Todorcevic:

**Theorem 2.1** (Kechris–Solecki–Todorcevic [8, Proposition 4.6])**.** *If $G$ is a Borel graph of finite maximum degree $d$, then $\chi_{\mathrm{B}}(G) \leqslant d+1$.*

Fix a countable group $\Gamma$ and a finite subset $F \subseteq \Gamma$. Without loss of generality, we may assume that $\mathbf{1} \notin F$. Say that a set $I \subseteq 2^\Gamma$ is *independent* if $I \cap (\sigma \cdot I) = \varnothing$ for all $\sigma \in F$ (when $I \subseteq \mathrm{Free}(2^\Gamma)$, this exactly means that $I$ is $G(\mathrm{Free}(2^\Gamma), F)$-independent). For brevity, let

$$\alpha_\beta \coloneqq \alpha_\beta\left(G\left(\mathrm{Free}\left(2^\Gamma\right), F\right)\right).$$

**Lemma 2.2.** *For every $\alpha < \alpha_\beta$, there is a clopen independent set $I \subseteq 2^\Gamma$ such that $\beta(I) \geqslant \alpha$.*

*Proof.* Let $J \subseteq \mathrm{Free}(2^\Gamma)$ be a $\beta$-measurable independent set with $\beta(J) > \alpha$. Since $\beta$ is regular [6, Theorem 17.10] and $2^\Gamma$ is zero-dimensional, there is a clopen set $C \subseteq 2^\Gamma$ with

$$\mu(J \mathbin{\triangle} C) \leqslant \frac{\beta(J) - \alpha}{|F|+1}.$$

Set $I \coloneqq C \setminus \bigcup_{\sigma\in F}(\sigma \cdot C)$. By construction, $I$ is clopen and independent. Moreover, if $x \in J \setminus I$, then either $x \in J \setminus C$ or $x \in (\sigma \cdot C) \setminus (\sigma \cdot J)$ for some $\sigma \in F$. Therefore,

$$\beta(I) \geqslant \beta(J) - (|F|+1)\beta(J \mathbin{\triangle} C) \geqslant \alpha. \qquad \square$$

Let $\Gamma \curvearrowright X$ be a free Borel action on a standard Borel space. Fix an arbitrary clopen independent set $I \subseteq 2^\Gamma$. We will prove that $\chi^*_{\mathrm{B}}(G(X,F)) \leqslant 1/\beta(I)$, which yields Theorem 1.1 by Lemma 2.2. Since $I$ is clopen, there exist finite sets $D \subseteq \Gamma$ and $\Phi \subseteq 2^D$ such that

$$I = \left\{x \in 2^\Gamma : x|_D \in \Phi\right\},$$

where $x|_D$ denotes the restriction of $x$ to $D$. Note that

$$\beta(I) = \frac{|\Phi|}{2^{|D|}}.$$

Let $N \coloneqq |DD^{-1}|$ and consider the graph $H \coloneqq G(X, DD^{-1})$. Every vertex in $H$ has precisely $N-1$ neighbors (we are subtracting 1 to account for the fact that a vertex is not adjacent to itself). By Theorem 2.1, this implies that $\chi_{\mathrm{B}}(H) \leqslant N$. In other words, we may fix a Borel function $f\colon X \to N$ such that $f(u) \neq f(v)$ whenever $u$, $v \in X$ are distinct points satisfying $v \in DD^{-1} \cdot u$. This implies that for each $x \in X$, the restriction of $f$ to the set $D \cdot x$ is injective. Now, to each mapping $\varphi\colon N \to 2$, we associate a Borel $\Gamma$-equivariant function $\pi_\varphi\colon X \to 2^\Gamma$ as follows:

$$\pi_\varphi(x)(\gamma) \coloneqq (\varphi \circ f)(\gamma \cdot x) \quad \text{for all} \quad x \in X \quad \text{and} \quad \gamma \in \Gamma.$$

Let $I_\varphi \coloneqq \pi_\varphi^{-1}(I)$. Since $\pi_\varphi$ is $\Gamma$-equivariant, $I_\varphi$ is $G(X,F)$-independent. Consider any $x \in X$ and let

$$S_x \coloneqq \{f(\gamma \cdot x) : \gamma \in D\}.$$

By the choice of $f$, $S_x$ is a subset of $N$ of size $|D|$. Whether or not $x$ is in $I_\varphi$ is determined by the restriction of $\varphi$ to $S_x$; furthermore, there are exactly $|\Phi|$ such restrictions that put $x$ in $I_\varphi$. Thus, the number of mappings $\varphi\colon N \to 2$ for which $x \in I_\varphi$ is

$$|\Phi| 2^{N-|D|} = \beta(I) 2^N.$$

Since this holds for all $x \in X$, we conclude that the sets $I_\varphi$ cover every point in $X$ exactly $\beta(I)2^N$ times. Therefore, $\chi^*_{\mathrm{B}}(G(X,F)) \leqslant 1/\beta(I)$, as desired.

## 3. Proof of Corollary 1.4

Thanks to Theorem 1.1, in order to establish Corollary 1.4 it is enough to verify that

$$\alpha_\beta\left(G_n\right) = \left(\frac{1}{2} + o(1)\right)\frac{\log n}{n}.$$

There are a number of known constructions that witness the lower bound

$$\alpha_\beta\left(G_n\right) \geqslant \left(\frac{1}{2} + o(1)\right)\frac{\log n}{n};$$

see, e.g., [9] by Lauer and Wormald and [4] by Gamarnik and Goldberg. Moreover, by [2, Corollary 1.2], even the inequality $\chi_\beta(G_n) \leqslant (2+o(1))n/\log n$ holds. For the upper bound

$$\alpha_\beta(G_n) \leqslant \left(\frac{1}{2} + o(1)\right)\frac{\log n}{n}, \tag{3}$$

we shall use a theorem of Rahman and Virág [13], which says that the largest density of a factor of i.i.d. independent set in the $d$-regular tree is at most $(1+o(1))\log d/d$. In the remainder of this section we describe their result and explain how it implies the desired upper bound on $\alpha_\beta(G_n)$.

Fix an integer $n \geqslant 1$. For our purposes, it will be somewhat more convenient to work on the space $[0,1]^{\mathbb{F}_n}$ instead of $2^{\mathbb{F}_n}$, where $[0,1]$ is the unit interval equipped with the usual Lebesgue probability measure. The product measure on $[0,1]^{\mathbb{F}_n}$ is denoted by $\lambda$. Let $H_n$ be the Schreier graph of the shift action $\mathbb{F}_n \curvearrowright [0,1]^{\mathbb{F}_n}$ corresponding to the standard generating set of $\mathbb{F}_n$. We remark that, by a theorem of Abért and Weiss [1] (see also [7, Theorem 6.46]), $\alpha_\beta(G_n) = \alpha_\lambda(H_n)$, so it does not really matter whether we are working with $G_n$ or $H_n$.

Set $d \coloneqq 2n$ and let $\mathbb{T}_d$ denote the Cayley graph of the free group $\mathbb{F}_n$ with respect to the standard generating set. In other words, $\mathbb{T}_d$ is an (infinite) $d$-regular tree. We view $\mathbb{T}_d$ as a *rooted* tree, whose root is the vertex $\mathbf{1}$, i.e., the identity element of $\mathbb{F}_n$. Let $\mathfrak{A}$ be the automorphism group of $\mathbb{T}_d$, i.e., the set of all bijections $\mathfrak{A}\colon \mathbb{F}_n \to \mathbb{F}_n$ that preserve the edges of $\mathbb{T}_d$, and let $\mathfrak{A}_\bullet \subseteq \mathfrak{A}$ be the subgroup comprising the root-preserving automorphisms, i.e., those $\mathfrak{A} \in \mathfrak{A}$ that map $\mathbf{1}$ to $\mathbf{1}$. The space $[0,1]^{\mathbb{F}_n}$ is equipped with a natural right action $[0,1]^{\mathbb{F}_n} \circlearrowleft \mathfrak{A}$. Namely, for $\mathfrak{A} \in \mathfrak{A}$ and $x \in [0,1]^{\mathbb{F}_n}$, the result of acting by $\mathfrak{A}$ on $x$ is the function $x \cdot \mathfrak{A}\colon \mathbb{F}_n \to [0,1]$ given by

$$(x \cdot \mathfrak{A})(\delta) \coloneqq x(\mathfrak{A}(\delta)) \quad \text{for all} \quad \delta \in \mathbb{F}_n.$$

For each $\gamma \in \mathbb{F}_n$, there is a corresponding automorphism $\mathfrak{A}_\gamma \in \mathfrak{A}$ sending every group element $\delta \in \mathbb{F}_n$ to $\delta\gamma$. The mapping $\mathbb{F}_n \to \mathfrak{A}\colon \gamma \mapsto \mathfrak{A}_\gamma$ is an antihomomorphism of groups, that is, we have

$$\mathfrak{A}_{\gamma\sigma} = \mathfrak{A}_\sigma \circ \mathfrak{A}_\gamma \quad \text{for all} \quad \gamma, \sigma \in \mathbb{F}_n,$$

where $\circ$ denotes composition. In particular, $\{\mathfrak{A}_\gamma : \gamma \in \mathbb{F}_n\}$ is a subgroup of $\mathfrak{A}$ isomorphic to $\mathbb{F}_n$. The right action $[0,1]^{\mathbb{F}_n} \circlearrowleft \mathfrak{A}$ and the left action $\mathbb{F}_n \curvearrowright [0,1]^{\mathbb{F}_n}$ are related by the formula

$$x \cdot \mathfrak{A}_\gamma = \gamma \cdot x \quad \text{for all} \quad x \in [0,1]^{\mathbb{F}_n}.$$

A set $X \subseteq [0,1]^{\mathbb{F}_n}$ is called $\mathfrak{A}_\bullet$-*invariant* if $x \cdot \mathfrak{A} \in X$ for all $x \in X$ and $\mathfrak{A} \in \mathfrak{A}_\bullet$. The Rahman–Virág theorem can now be stated as follows:

**Theorem 3.1** (Rahman–Virág [13, Theorem 2.1])**.** *If $I \subseteq [0,1]^{\mathbb{F}_n}$ is an $\mathfrak{A}_\bullet$-invariant $\lambda$-measurable $H_n$-independent set, then*

$$\lambda(I) \leqslant (1+o(1))\frac{\log d}{d} = \left(\frac{1}{2}+o(1)\right)\frac{\log n}{n}.$$

Theorem 3.1 is almost the result we want, except that we need an upper bound on the measure of *every* (not necessarily $\mathfrak{A}_\bullet$-invariant) $\lambda$-measurable $H_n$-independent set $I$. To remove the $\mathfrak{A}_\bullet$-invariance assumption, we use the following consequence of Theorem 3.1:

**Corollary 3.2.** *There exists a Borel graph $Q$ with a probability measure $\mu$ on $V(Q)$ such that:*

- *every connected component of $Q$ is a $d$-regular tree; and*
- *$\alpha_\mu(Q) \leqslant (1/2+o(1))\log n/n$.*

*Proof.* We use a construction that was studied by Conley, Kechris, and Tucker–Drob in [3]. Let $\Omega$ be the set of all points $x \in [0,1]^{\mathbb{F}_n}$ such that $x \cdot \mathfrak{A} \neq x$ for every non-identity automorphism $\mathfrak{A} \in \mathfrak{A}$. Let us make a couple observations about $\Omega$. Notice that, by definition, the set $\Omega$ is invariant under the action $[0,1]^{\mathbb{F}_n} \circlearrowleft \mathfrak{A}$; in particular, it is invariant under the shift action $\mathbb{F}_n \curvearrowright [0,1]^{\mathbb{F}_n}$. Furthermore, the induced action of $\mathbb{F}_n$ on $\Omega$ is free (indeed, even the action $\Omega \circlearrowleft \mathfrak{A}$ is free). Since every injective mapping $\mathbb{F}_n \to [0,1]$ belongs to $\Omega$, we conclude that $\lambda(\Omega) = 1$. Now consider the quotient space $V \coloneqq \Omega/\mathfrak{A}_\bullet$. As the group $\mathfrak{A}_\bullet$ is compact, the space $V$ is standard Borel [3, paragraph preceding Lemma 7.8]. Let $\mu$ be the push-forward of $\lambda$ under the quotient map $\Omega \to V$, and let $Q$ be the graph with vertex set $V$ in which two vertices $\boldsymbol{x}, \boldsymbol{y} \in V$ are adjacent if and only if there are representatives $x \in \boldsymbol{x}$ and $y \in \boldsymbol{y}$ that are adjacent in $H_n$. Conley, Kechris, and Tucker–Drob [3, Lemma 7.9] (see also [16, Proposition 1.9]) showed that every connected component of $Q$ is a $d$-regular tree. Furthermore, by construction, a set $I \subseteq V$ is $Q$-independent if and only if its preimage under the quotient map is $H_n$-independent. Since the quotient map establishes a one-to-one correspondence between subsets of $V$ and $\mathfrak{A}_\bullet$-invariant subsets of $\Omega$, Theorem 3.1 is equivalent to the assertion that $\alpha_\mu(Q) \leqslant (1/2+o(1))\log n/n$, as desired. □

In view of Corollary 3.2, the following lemma completes the proof of (3):

**Lemma 3.3.** Let $Q$ be a Borel graph in which every connected component is a $d$-regular tree and let $\mu$ be a probability measure on $V(Q)$. Then $\alpha_\mu(Q) \geqslant \alpha_\beta(G_n)$.

In the case when $Q$ is the Schreier graph of a free measure-preserving action of $\mathbb{F}_n$, the conclusion of Lemma 3.3 follows from the Abért–Weiss theorem [1]. To handle the general case, we rely on a strengthening of a recent result of Tóth [17] due to Grebík [5], which, roughly, asserts that every $d$-regular Borel graph is "approximately" induced by an action of $\mathbb{F}_n$.

To state this result precisely, we introduce the following terminology. A *Borel partial action* $\boldsymbol{p}$ of $\mathbb{F}_n$ on a standard Borel space $X$, in symbols $\boldsymbol{p}\colon \mathbb{F}_n \curvearrowright^* X$, is a sequence of Borel partial injections $p_1, \dots, p_n\colon X \dashrightarrow X$. Given a Borel graph $Q$, we say that a Borel partial action $\boldsymbol{p}\colon \mathbb{F}_n \curvearrowright^* V(Q)$ is a *partial Schreier decoration* of $Q$ if $p_i(x)$ is adjacent to $x$ for all $1 \leqslant i \leqslant n$ and $x \in \mathrm{dom}(p_i)$. If $\boldsymbol{p}$ is a partial Schreier decoration of a graph $Q$, then we let $C(Q, \boldsymbol{p})$ be the set of all vertices $x \in V(Q)$ such that $x$ belongs to both the domain and the image of every $p_i$ and the neighborhood of $x$ in $Q$ is equal to the set $\{p_1(x), \dots, p_n(x), p_1^{-1}(x), \dots, p_n^{-1}(x)\}$. A *Schreier decoration* of $Q$ is a partial Schreier decoration $\boldsymbol{p}$ such that $C(Q, \boldsymbol{p}) = V(Q)$. It is easy to see that $Q$ admits a Schreier decoration if and only if it is the Schreier graph of a Borel action of $\mathbb{F}_n$.

Now we can state Grebík's result:

**Theorem 3.4** (Grebík [5, Theorem 0.2(III)])**.** *Let $Q$ be a $d$-regular Borel graph and let $\mu$ be a probability measure on $V(Q)$. Then for every $\varepsilon > 0$, $Q$ admits a partial Schreier decoration $\boldsymbol{p}$ such that $\mu(C(Q, \boldsymbol{p})) \geqslant 1 - \varepsilon$.*

With Theorem 3.4 in hand, we are ready to establish Lemma 3.3.

*Proof of Lemma 3.3.* Recall that we denote the generators of $\mathbb{F}_n$ by $\sigma_1, \dots, \sigma_n$. Let $Q$ be a Borel graph in which every connected component is a $d$-regular tree and let $\mu$ be a probability measure on $V(Q)$. Thanks to Lemma 2.2, it suffices to show that $\alpha_\mu(Q) \geqslant \beta(I)$ for every clopen independent set $I \subseteq 2^{\mathbb{F}_n}$, where, as in § 2, we say that $I$ is independent if $I \cap (\sigma_i \cdot I) = \varnothing$ for each $1 \leqslant i \leqslant n$.

Fix a clopen independent set $I \subseteq 2^{\mathbb{F}_n}$. Since $I$ is clopen, we can write

$$I = \left\{x \in 2^{\mathbb{F}_n} : x|_D \in \Phi\right\},$$

where $D \subset \mathbb{F}_n$ and $\Phi \subseteq 2^D$ are finite sets. Furthermore, we may assume without loss of generality that $D = \{\gamma \in \mathbb{F}_n : |\gamma| \leqslant k\}$ for some $k \in \mathbb{N}$, where $|\gamma|$ denotes the word norm of $\gamma$. For a vertex $x \in V(Q)$, we let $N^k(x)$ be the set of all vertices that are joined to $x$ by a path of length at most $k$. Since every connected component of $Q$ is a $d$-regular tree, we have $|N^k(x)| = |D|$ for all $x \in V(Q)$. This allows us to define a probability measure $\mu_k$ on $V(Q)$ via

$$\mu_k(A) \coloneqq \int \frac{\left|A \cap N^k(x)\right|}{|D|}\,\mathrm{d}\mu(x) \quad \text{for all Borel } A \subseteq V(Q).$$

We have now prepared the ground for an application of Theorem 3.4. Fix $\varepsilon > 0$ and let $\boldsymbol{p}$ be a partial Schreier decoration of $Q$ such that

$$\mu_k(C(Q, \boldsymbol{p})) \geqslant 1 - \frac{\varepsilon}{|D|},$$

which exists by Theorem 3.4. Let $C_k$ be the set of all $x \in V(Q)$ such that $N^k(x) \subseteq C(Q, \boldsymbol{p})$. Then

$$\begin{aligned} 1 - \frac{\varepsilon}{|D|} &\leqslant \mu_k\left(C\left(Q, \boldsymbol{p}\right)\right) \\ &= \int \frac{\left|C\left(Q, \boldsymbol{p}\right) \cap N^k(x)\right|}{|D|}\,\mathrm{d}\mu(x) \leqslant \mu(C_k) + \left(1 - \frac{1}{|D|}\right)\left(1 - \mu\left(C_k\right)\right) \\ &= \frac{1}{|D|}\mu(C_k) + 1 - \frac{1}{|D|}, \end{aligned}$$

which implies that $\mu(C_k) \geqslant 1 - \varepsilon$. The importance of the set $C_k$ lies in the fact that for each $x \in C_k$ and $\gamma \in D$, there is a natural way to define the notation $\gamma \cdot x$. Namely, we write $\gamma$ as a reduced word:

$$\gamma = \sigma_{i_1}^{s_1} \cdots \sigma_{i_\ell}^{s_\ell},$$

where $0 \leqslant \ell \leqslant k$, each index $i_j$ is between 1 and $n$, and each $s_j$ is 1 or $-1$. Since $N^k(x) \subseteq C(Q, \boldsymbol{p})$, there is a unique sequence $x_0$, $x_1$, …, $x_\ell$ of vertices with

$$x_0 = x \quad \text{and} \quad x_j = p_{i_j}^{s_j}\left(x_{j-1}\right) \text{ for all } 1 \leqslant j \leqslant \ell.$$

We then set $\gamma \cdot x \coloneqq x_\ell$. Note that we have $N^k(x) = \{\gamma \cdot x : \gamma \in D\}$.

The remainder of the argument utilizes a construction similar to the one in the proof of Theorem 1.1 given in § 2. Consider the graph $R$ with the same vertex set as $Q$ in which two distinct vertices are adjacent if and only if they are joined by a path of length at most $2k$ in $Q$. Since every connected component of $Q$ is a $d$-regular tree, each vertex in $R$ has the same finite number of neighbors, so, by Theorem 2.1, the

Borel chromatic number $\chi_{\mathrm{B}}(R)$ is finite. Let $N \coloneqq \chi_{\mathrm{B}}(R)$ and fix a Borel function $f\colon V(Q) \to N$ such that $f(u) \neq f(v)$ whenever $u$ and $v$ are adjacent in $R$. Then for each $x \in V(Q)$, the restriction of $f$ to the set $N^k(x)$ is injective. Now, to each mapping $\varphi\colon N \to 2$, we associate function $\pi_\varphi\colon C_k \to 2^D$ as follows:

$$\pi_\varphi(x)(\gamma) \coloneqq (\varphi \circ f)(\gamma \cdot x) \quad \text{for all } x \in C_k \text{ and } \gamma \in D.$$

Let $I_\varphi \coloneqq \{x \in C_k \,:\, \pi_\varphi(x) \in \Phi\}$. The independence of $I$ implies that the set $I_\varphi$ is $Q$-independent. We will show that for some choice of $\varphi\colon N \to 2$, $\mu(I_\varphi) \geqslant (1-\varepsilon)\beta(I)$. Since $\varepsilon$ is arbitrary, this yields the desired bound $\alpha_\mu(Q) \geqslant \beta(I)$ and completes the proof of Lemma 3.3.

Consider any $x \in C_k$ and let

$$S_x \coloneqq \{f(\gamma \cdot x) \,:\, \gamma \in D\}.$$

Since $f$ is injective on $N^k(x)$, $S_x$ is a subset of $N$ of size $|D|$. Whether or not $x$ is in $I_\varphi$ is determined by the restriction of $\varphi$ to $S_x$; furthermore, there are exactly $|\Phi|$ such restrictions that put $x$ in $I_\varphi$. Thus, the number of mappings $\varphi\colon N \to 2$ for which $x \in I_\varphi$ is

$$|\Phi| 2^{N-|D|} \,=\, \beta(I) 2^N.$$

Since this holds for all $x \in C_k$, we conclude that

$$\sum_{\varphi\,:\,N \to 2} \mu(I_\varphi) \,\geqslant\, \mu\left(C_k\right) \beta(I) 2^N \,\geqslant\, (1-\varepsilon)\beta(I) 2^N,$$

where the second inequality uses that $\mu(C_k) \geqslant 1-\varepsilon$. In other words, the average value of $\mu(I_\varphi)$ over all $\varphi\colon N \to 2$ is at least $(1-\varepsilon)\beta(I)$. Thus, the maximum is at least $(1-\varepsilon)\beta(I)$ as well, and the proof is complete. □

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